

Hugo Folloni Guarilha

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Professional Summary

Full-stack software engineer with 3+ years of experience building robust, scalable applications. Expertise in tackling complex data challenges using React, .NET, Python, and Docker to develop critical systems for educational platforms, government institutions, and technology companies. Backed by a Computer Science degree from UFRJ and C2 professional English proficiency.

Skills

- JavaScript | TypeScript | React | C# | .NET | Python | Django | Node | Express | Redux | Next.js | PostgreSQL | SQL Server | MongoDB | Git
- Frontend | Backend | Full-Stack | SQL | NoSQL | AWS | Docker | Scrum | Agile | English and Portuguese – *Professional proficiency or above*

Experience

Jr. Software Engineer BigDataCorp *Rio de Janeiro, Brazil* **01/2026 - present**

- Orchestrated the end-to-end development of core billing and monetization platforms (**React**, **.NET**), managing complex data flows for contracts, invoices, and accounts receivable to generate consolidated financial reports for leadership and operations.
- Engineered the transition to a unified Price Tables model, successfully migrating data for over 5,000 enterprise clients and 10,000 contracts to support 9 product lines and scale the billing ecosystem.
- Implemented a proactive caching layer that reduced data retrieval times by up to 90%, and achieved 100% automated test coverage for the core API by establishing internal **TDD** protocols, using **xUnit**.

Software Engineer Intern BigDataCorp *Rio de Janeiro, Brazil* **06/2024 - 12/2025**

- Developed key features for internal administrative panels, architecting **RESTful** and **event-driven** APIs (**.NET 8**) and responsive (**React**) components to enable critical financial and selling data management and streamline workflows for 50+ internal users.
- Built and deployed automated Purchase Order (PO) tracking systems and credit workflows, replacing manual Omie ERP checks and directly accelerating daily operations for the Finance and Sales teams, integrating **AWS S3** document saving.
- Spearheaded a major stack modernization effort, upgrading legacy backend APIs to **.NET 8** and migrating essential User Interface components to a **React 18** architecture.

Scientific Research Fellow - ETL COPPETEC *Rio de Janeiro, Brazil* **06/2023 - 12/2025**

- Designed and deployed 10 new **Talend ETL** workflows and maintained a larger 40-job data integration architecture, securely processing legal and institutional data using **PostgreSQL** and **Docker**.
- Executed zero-downtime database operations and targeted data corrections across production **PostgreSQL** and **SQL Server** environments for a major public institution, strictly maintaining data integrity and operational reliability for critical state systems.
- Led the development of solutions for workspace sharing, creating a method to host the **Talend** application in **Docker containers**.

Scientific Research Fellow - Development Machine Teaching *Rio de Janeiro, Brazil* **02/2022 - 12/2025**

- Led the development of a synchronous assessment module on a major Computer Science educational platform as my capstone thesis project, unifying the platform's evaluation cycle within a secure, time-controlled exam environment.
- Optimized user data dashboards by rewriting database queries, reducing load times by over 70%.
- Automated backend and User Interface testing suites with **Playwright** and **DjangoTesting**, establishing a strong foundation for platform reliability.

Education

Bachelor of Science Universidade Federal do Rio de Janeiro *Rio de Janeiro, Brazil* **02/2021 - 11/2025**

- Major in **Computer Science** (Top-ranked university in Brazil, recognized for excellence in STEM fields)
Capstone Thesis: "Implementation of Synchronous Assessments on the Machine Teaching Platform"

Projects

- **vinyl-scrobbler**: Engineered a full-stack application (**React 19**, **TypeScript**, **.NET 9**) to digitize physical vinyl playback, integrating the Alexa Skills Kit for voice-activated scrobbling and utilizing Spotify/Last.fm APIs for automated metadata retrieval and timestamp calculations.
- **ye.bio**: A full-stack 'link-in-bio' creator with customizable layouts, built with **React**, **C#**, **.NET**, and **PostgreSQL**.
- **spotify-multiverse**: A music discovery app (**React 18**, **Python**) that generates personalized recommendations by applying Linear Algebra to playlist attributes.

Freelancer

- **Instituto de Proteção ao Jogador:** Developed the complete system (**React, Node.js, PostgreSQL**) for a psychological project aimed at helping individuals with gambling addiction, in partnership with Brazilian betting companies.
- **Bracell - Floresta Sempre Viva:** Developed an interactive website (**React**) for an environmental project, in association with a major company.

Publications & Presentations

- **Author – WEI 2026 (Brazilian Computer Society Congress):** *"Beyond Correctness: A Competency-Driven Framework for Designing Autograder Test Suites"* (DOI: 10.5753/wei.2026.21827).
- **Co-author** — Revista Brasileira de Informática na Educação (RBIE), 2025: "Technical and Tactical Factors of an Educational Platform for a Large Number of Students" (DOI: 10.5753/rbie.2025.4479) - Award-Winning Article.
- **Presenter** — Semana de Integração Acadêmica (SIAC UFRJ), 2024: Received **Best Presentation Award** for the study "Testes de Servidor no Machine Teaching"
- **Presenter** — Semana de Integração Acadêmica (SIAC UFRJ), 2025: Received **Best Presentation Award** for the study "Funcionalidade de Avaliação Síncrona na Plataforma Machine Teaching"

Others

- **Certificate of English Language Proficiency C2** – The highest-level certificate of English proficiency (**02/2025**)